

Question for a Uni Design Project: Is the massive energy footprint of AI actually on your radar?

Posted by u/TalNix77 • • 0 points • 17 comments

Hi everyone,

I'm a design student researching the "invisible" energy consumption of AI for a university project.

While the utility of tools like ChatGPT is obvious, the physical resources required to run them are massive. Studies suggest that a single generative AI query can consume significantly more energy than a standard web search (some estimates range from 10x to 25x more).

I'm looking for honest perspectives on this:

1. **Awareness:** Before reading this, were you actually aware of the scale of energy difference between a standard search and an AI prompt? Or is that completely "invisible" in your daily usage?
2. **Impact on Usage:** Does the energy intensity play any role in how you use these tools? Or is the utility simply the only factor that matters for your workflow?
3. **Value vs. Waste:** Do you view this high energy consumption as a fair investment for the results you get, or does the current technology feel inefficient to you?

I'm trying to get a realistic picture of whether this topic actually plays a role in users' minds or if performance is the priority.

Comments

u/teapot_RGB_color • 1 points • 2025-12-06 03:58 UTC

Also to be noted. Your reference is wrong.

A standard search does not provide you with an answer. Only redirect you to where you can find an answer, e.g. YouTube.

So is like to know the energy consumption of what that might look like.

u/teapot_RGB_color • 2 points • 2025-12-06 03:50 UTC

Energy usage of social media is way more massive than AI. No one cares.

Energy usage of crypto is extreme. In addition the energy is pure waste /discard.

Is like to know what makes you think AI is special?

u/rakuu • 0 points • 2025-12-06 01:59 UTC

You should research this more. 10x more energy than a Google search isn't that much, and if that was the issue, it would not be much of an issue, except in local areas around new data centers to serve inference (mostly in the global south) - but that was an issue that existed for all data centers and isn't unique to AI. The data center energy needs to watch YouTube or Netflix for example are much greater.

Most of the energy used by data centers is used for training and research, months before any end user types something into that ChatGPT or Gemini model. This is what is needed to be understood to understand the larger energy impact.

u/Titanium-Marshmallow • 2 points • 2025-12-06 01:34 UTC

We need honest, impartial, accurate datasets on power consumption. Energy was a major problem before this heh. Yea things are getting so much better, sure, right

I don't want to go political so I'll just say this is a critical problem and credible data is crucial.

What's a truly authoritative source for metrics on power consumption per unit (time, query, user x time, anything)

Aware: yes. I try to only use what I need and not be wasteful, like with all that I do.

u/ForeignAdvantage5198 • 0 points • 2025-12-06 00:41 UTC

should it be?

u/Brighter-Side-News • 1 points • 2025-12-06 00:30 UTC

We've written quite extensively about the massive energy requirements for AI and the sustainable options to meet future demand.

u/NerdyWeightLifter • 1 points • 2025-12-06 00:01 UTC

I'm quite aware of the relative energy use, but this is a rapidly moving target.

The per-unit-cognition cost has been dropping by around 70% per annum for several years now.

The first few years of these gains we soaked up by offering the same priced AI services, but with more cognition. Have you noticed GPT 5 "thinking" mode running for several minutes on big problems?

We're pushing the limits on typical user demand for more cognition per query, so now they're pushing into Agentic AI, where it's given long term goals.

Meanwhile, businesses are working through the initially slow process of AI integration into all of their business processes. This just gets faster with time though. It snowballs in a feedback loop.

There are also various major step functions in terms of AI processing efficiency ahead of us. Neuromorphic chips, photonic chips, thermodynamic computing (lookup "Extropic"), and eventually quantum computing.

Comparing this to search queries, is far too simplistic.

u/HeresyClock • 3 points • 2025-12-05 23:27 UTC

Datacenters and computing use enormous amount of energy, that is true and should raise concern. However, however... chatgpt prompts are a drop in the ocean in the whole consumption and focusing on that is frankly idiotic and counter productive. Like giving the impression that cutting back on chatgpt queries would be the saving grace and end of concern.

You can't even google without engaging the embedded AI helper, has that been calculated on the 10x to 25x?

Watching Netflix for an hour takes 0.12 kWh. That is 500 Gemini prompts. 500!!! For a single hour. And yet, who bats an eye if people binge Netflix for hours?

Figure out where the energy and resources are really used and target those. Model training and development, and private model use, as well as video and image creation are good guesses. (Video generation is resource heavy, for hour of Netflix is maybe 20s of video.)

Source (first one that came up on google, and there is lot of similar info available)

<https://www.forbes.com/sites/johnkoetsier/2025/12/03/new-data-ai-is-almost-green-compared-to-netflix-zoom-youtube/>

My answers: let me ponder on my terrible energy usage of asking chatbot instead of google while I watch the new Stranger Things... no, sorry. A real answer: both text AI and google query use such tiny amounts of energy that it makes no difference. If I went all out and made 100 AI queries a day, in a month that would be equal to 6 hours of Netflix. Am I worried about watching Netflix for six hours in a month? No. So why would I be worried over AI?

What I am concerned about are the datacenters, and the whole of it.

u/Ciappatos • 1 points • 2025-12-05 23:18 UTC

1. Yes.

2. Yes.

3. The latter.

"While the utility of tools like ChatGPT is obvious" This statement is far from obvious and I don't think you should state it as a fact.

u/HumanInTheLoop30 • 0 points • 2025-12-05 23:05 UTC

Here's my perspective based on a very concrete experience.

I've been working with an AI assistant daily for about a month on a system-level design project. One thing I noticed — almost by accident — is that the "energy footprint" of AI becomes very visible

when using it on mobile.

Not because mobile is special, but because the device itself reacts physically:

- When the AI runs deeper reasoning chains, the phone heats up.
- Battery drains rapidly.
- Response time slows down as the reasoning depth increases.
- And when I simplify the task or structure the prompt better, everything calms down.

This became a kind of real-time proxy for computational load — something you feel instead of just reading about in studies.

So here's how I'd answer your questions:

1. Awareness

I didn't know the exact numbers, but I felt the difference.

Heavy reasoning → heat + battery drain.

It made the "invisible cost" suddenly visible.

2. Impact on usage

Yes, it actually changed my workflow.

I started restructuring prompts to reduce the depth of reasoning layers (from ~8–10 layers down to ~4–5).

The AI responded faster, used fewer resources, and the phone stopped overheating.

So energy intensity indirectly shaped how I design my interactions.

3. Value vs. Waste

For me, the energy cost feels meaningful when the output has real value.

But if I force the AI into unnecessary complexity, it becomes obvious — not just conceptually, but physically — that the cost isn't worth it.

Most users never experience this because they work on desktops or cloud machines.

But mobile makes the energy curve "visible," and once you notice it, you naturally start optimizing your workflow.

u/Seishomin • 0 points • 2025-12-05 22:31 UTC

Yes I'm aware.

No it's not moderating my usage.

I'm seeing massive benefits - but most importantly the costs aren't visible to me in a significant way

u/LoLoL_the_Walker • 1 points • 2025-12-06 06:56 UTC

So you're saying you don't give a fuck as long as you're good?

u/Seishomin • 2 points • 2025-12-06 07:08 UTC

No I'm saying society won't give a fuck if the costs aren't visible

u/Comprehensive-Run615 • 2 points • 2025-12-05 21:43 UTC

Net net feels like more energy consumption, and need to push for more renewables sources of energy to compensate

Humans brains think at infinitely more energy efficiency than LLM today so it's already an unfair comparison on energy unit, we haven't quite figured it out yet the human brain.

In your example, what LLM enables is doing thousands x the number of searches a team of humans can do. thoroughness and speed. This generates value for a company because net net it generates more output for less \$cost. The \$cost is different tho since it's energy vs. human labor costs

So yes in this simple example if you look at AI replacing humans for mundane and data intensive tasks, it's a net energy consumption increase, but lower cost of doing business cus labor costs is definitely higher than energy.

Your benchmark example should not be a simple search since that's not really AI. Inference and judgement is the unlock.

u/Oona22 • 4 points • 2025-12-05 21:39 UTC

very much so, and am both perplexed and disturbed that government seems to be pushing so hard to use AI "for efficiencies" when 1. the error rate is INCREDIBLE and 2. it's atrocious for the environment

u/Extension-Two-2807 • 2 points • 2025-12-06 01:24 UTC

Yeah it's a pyramid scheme. It's a crappy search engine that talks to you and lies... constantly. It doesn't check sources and just regurgitates shit from all corners of the internet. I'm reminded of the commercial (forgive me I forget the product) but a man (conventionally unattractive), exclaims he's a French model to which the girl he's with (conventionally attractive) exclaims how it must be true because she read it on the internet. That girl, which was supposed to be a joke, is now pretty much the essence of AI. But what propels all the investments into AI? Greed. Greedy companies want to rid themselves of expensive employees and the AI grifters (also greedy) know this and so keep using these companies boundless greed to trick them into giving them more and more money for junk products with the promise that the real good one is JUST around the corner. Funny enough the inevitable long term (and even short term in my fortune 50 company) is a junk product that wastes time and is often used incorrectly by junk offshore workers (making their work output somehow even worse) which in turn creates a horrible customer experience and decimates the very economy many of them depend on to survive.

u/AutoModerator • 1 points • 2025-12-05 21:22 UTC

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- * Please provide links to back up your arguments.
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